Time Series Analysis of Nike, Inc. Stock Returns: An ARIMA Approach

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INTRODUCTION

Background

Nike, Inc. is one of the most recognizable companies in the global athletic apparel industry, making its stock an interesting subject for financial analysis. In this study, we analyze the behavior of Nike's stock prices using time series models.

Data Collection

Our data set contains the Nike stock prices from December 1980 to December 2025, nearly 45 years of data displayed by daily stock prices over that time span. As of the past 15 years, the Nike stock has had its ups and downs, peaking at prices of approximately $175 during 2021, but currently being at approximately $63, its' lowest price since 2018. The objective of this study is to model and forecast Nike stock behavior using time series methods, specifically ARIMA modeling on log returns.

We obtained our dataset from Kaggle, a website that encompasses thousands of datasets that are easily accessible for any kind of project. While some of these datasets are not reliable, we ensured that this collection of data was by comparing it to other records of the Nike stock price to ensure that their display of prices and dates was accurate and correct. We compared the dataset to those of Yahoo Finance, the NASDAQ, and NYSE, to which all of them backed up the relevance and reliability of our chosen dataset. A snapshot of our data in an Excel Workbook format is provided below:

Date	Open	High	Low	Close	Volume
2019-05-2	76.68713	76.68713	75.71709	75.81776	7181900
2019-05-2	76.28447	76.68713	75.50662	76.5407	5762500
2019-05-2	75.9459	76.26619	75.73543	76.13808	5237800
2019-05-2	75.49747	76.00994	75.03076	75.62559	9704200
2019-05-2	75.22293	76.1106	75.12227	75.18633	5999000
2019-05-2	75.81775	76.1289	74.24374	74.3078	7927300
2019-05-2	73.85939	73.99666	71.10488	72.17558	10116900
2019-05-3	72.22133	72.96258	71.93764	72.53247	6177800
2019-05-3	71.57808	71.81668	70.72465	70.78889	8891100

Our dataset, as shown above, provides the open, high, low, close, and volume of the stock price aligned with the date that it occurred in the time series. We will not be utilizing the high, low, or volume of the stock because it will not be used through the project for any step of the time series analysis or forecasting steps. Instead, we will only consider the closing price.

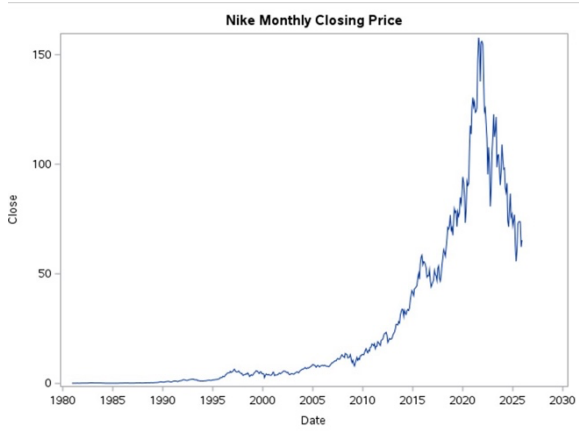
Through trial and error, we also decided to only include one value from every month of the data because of the thousands of observations of stock prices in the dataset. For the sake of simplicity, and due to the intensity of the observations and extensive computation time of using individual days, we will take the first day of each month associated with year and use that to forecast our prices one year into the future. The starting date of the data will also be December 1980 since an entire year of data is not available for 1980, due to the stock becoming public in November of 1980. In entirety, our data will now range from December 1980 to December 2025.

MODELS

For this dataset, we had the choice of analyzing either simple returns or log returns. For this analysis, the latter was used, particularly due to the additive properties over time and suitability of time series modeling.

Stationarity

We should note that in general, stock prices are non-stationary, while returns (particularly log returns) are often approximately stationary, therefore more suitable for modeling.



As we can see from the plot, there is a clear upward trend over time, along with increasing variability in later period. As previously mentioned, this behavior is normal for a financial price series, which is often followed by a random walk. Since time series models such as ARIMA require stationarity, the raw price data cannot be modeled without transformation.

Because of this, the series was transformed into log returns, which removed the trend and stabilized the variance:

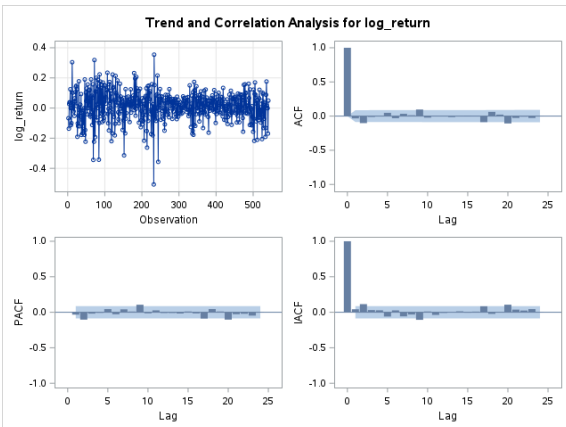
$$r_t = \log(P_t) - \log(P_{t-1})$$

Where

P_t : The stock price at time t.

r_t : The log return at time t.

After this transformation, the series fluctuates around a constant mean with no visible trend, indicating that the stationary assumption is satisfied. This indicates that the series is suitable for a time series modeling.



The ACF and PACF plots of the log returns show no significant autocorrelation, as nearly all lags fall within the confidence bounds. This suggests that the log return series behaves like white noise and does not exhibit a strong AR or MA structure.

Random Walk & White Noise

The behavior of the log return series is consistent with the random walk hypothesis, which suggests that stock price changes are unpredictable and can be driven by random shocks. Under the framework, prices follow a non-stationary process, while returns behave similarly to white noise.

For our AR(1), MA(1), and ARMA(1,1) models, our parameters were found as the following:

Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	0.01162	0.0040669	2.86	0.0045	0
AR1,1	-0.03349	0.04309	-0.78	0.4375	1

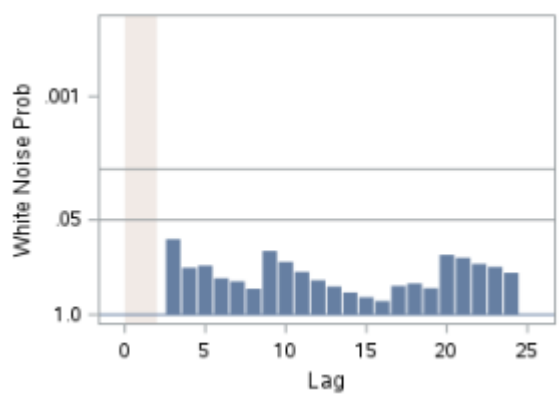
Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	0.01162	0.0040246	2.89	0.0040	0
MA1,1	0.04234	0.04307	0.98	0.3260	1

Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	0.01166	0.0036240	3.22	0.0014	0
MA1,1	0.57534	0.36553	1.57	0.1161	1
AR1,1	0.50806	0.38523	1.32	0.1878	1

Invertibility

We can see that the estimated AR and MA parameters were small and not statistically significant. This supports the conclusion that in the log return series, there is very little relationship between past returns and future returns, and the fitted model does not capture strong predictive structure.

Additionally, our series shows behavior close to white noise, therefore there does not exist any evidence of strong dependence that would violate the model's assumptions, and invertibility is not an issue in our model.

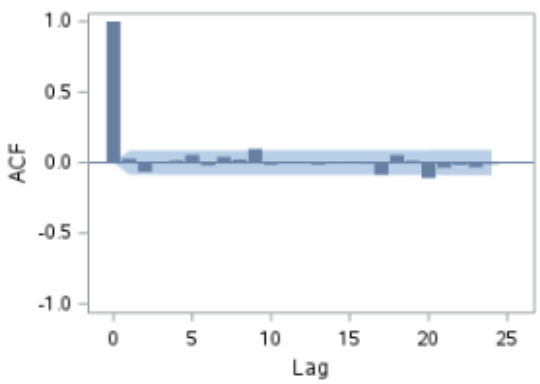


DIAGNOSTICS

To evaluate the adequacy of the fitted models, residuals diagnostics were performed.

The residuals from the ARMA(1,1) model were analyzed using autocorrelation functions and statistical tests. The results showed:

- Residual autocorrelations were within the confidence bounds.
- No significant spikes were observed in the residual ACF.

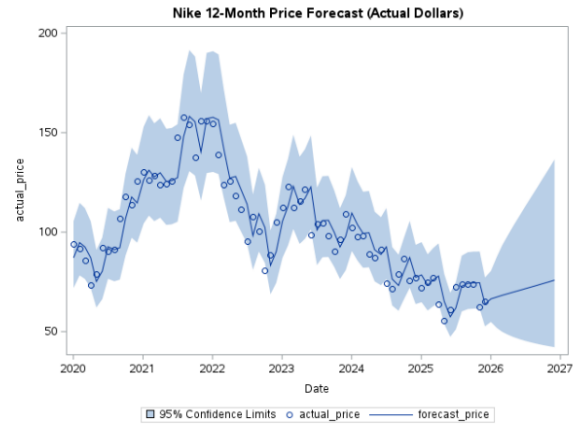


- Ljung-Box test p-values were all greater than 0.05 at multiple lags.

Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	4.72	4	0.3168	0.027	-0.067	0.004	0.015	0.053	-0.0
12	11.26	10	0.3378	0.039	0.021	0.098	-0.015	0.005	-0.0
18	17.19	16	0.3737	-0.014	-0.005	0.004	-0.007	-0.087	0.0
24	25.66	22	0.2666	0.013	-0.110	-0.036	-0.016	-0.033	-0.0
30	34.36	28	0.1893	-0.046	0.051	0.018	-0.066	0.061	0.0
36	44.96	34	0.0990	0.034	-0.030	-0.036	0.093	-0.024	-0.0
42	49.79	40	0.1381	0.024	0.018	-0.030	0.041	-0.019	0.0
48	51.16	46	0.2783	-0.015	-0.022	-0.004	-0.026	0.030	0.0

All these findings indicate that the residuals behave like white noise, meaning the model has indeed captured any linear dependence among the data.

FORECAST



The forecasted values for the next years' worth of Nike stock prices are shown in the "Nike 12-Month Price Forecast" plot. From this graph, we can conclude that our forecast, with all assumptions checking out for a unique and adequate model, that the Nike stock price over the next year will increase slightly.

The forecasted increase is not anything to squander, but it is not a significant increase in the stock price like we saw in 2020. This forecast more so seems that the price will rise back to the 2025 closing high of approximately $76.99 from its stock price at the end of the observed data of $65.39, a 17.74% increase.

However, we should take into consideration the fact that the confidence intervals widen over time, indicating great uncertainty in these predictions. Given the weak dependence structure and the behavior of the series being white noise, this forecast should be looked at with wariness, as it does not provide strong evidence of reliable predictability.

CONCLUSION

In this study, we analyzed Nike's stock price behavior using time series methods over a 45-year period. The original closing price was found to be non-stationary, and after transforming the data into log returns, the series shows signs of a stationary process, making it fit for ARIMA modeling.

Although the ARMA(1,1) model was best suited out of the three based on AIC, the estimated AR and MA parameters were not statistically significant. Additionally, diagnostic checks show that the residuals behave like white noise.

While the forecast results showed a slight upward movement in stock price, the (relatively) wide confidence intervals indicate a high level of uncertainty. We shall warn to interpret the forecast with caution, as it does not provide a stable level of evidence of predictability.

Overall, our results are consistent with the random walk hypothesis, which suggests the stock price changes are largely unpredictable and driven by random shocks.

Works Cited

https://www.kaggle.com/datasets/isaaclopgu/nike-stock-data-daily-updated?resource=download

https://finance.yahoo.com/quote/NKE/

SAS Code

```
proc import datafile="/home/u64137575/sasuser.v94/Nike_first_day_each_month.xlsx"
   out=work.nike_raw
   dbms=xlsx
   replace;
   sheet="in";
   getnames=yes;
run;


data work.nike_ts;
   set work.nike_raw;

   Date_only = scan(Date, 1, ' ');
   SAS_Date = input(Date_only, yymmdd10.);
   format SAS_Date yymmdd10.;

   keep SAS_Date Close Open High Low Volume ticker;
   rename SAS_Date = Date;
run;

proc sgplot data=work.nike_ts;
   series x=Date y=Close;
   xaxis type=time;
   title "Nike Monthly Closing Price";
run;
```

```
data work.nike_ts;
   set work.nike_ts;
   log_close = log(Close);
   log_return = log(Close) - log(lag(Close));
run;

proc sgplot data=work.nike_ts;
   series x=Date y=log_return;
   xaxis type=time;
   title "Nike Monthly Log Returns";
run;

proc arima data=work.nike_ts;
   identify var=log_return nlag=24 scan;
estimate p=1;
estimate q=1;
estimate p=1 q=1;
run;
quit;
data work.nike_final_forecast; set work.nike_forecast_results;
/* Exponentiate to get back to original price scale */
forecast_price = exp(FORECAST);
lower_price    = exp(L95);
upper_price    = exp(U95);
actual_price   = exp(log_close); /* recovers original Close price */

keep Date actual_price forecast_price lower_price upper_price;
```

```
run;

proc sgplot data=work.nike_final_forecast;

band x=Date lower=lower_price upper=upper_price / transparency=0.5 legendlabel="95% CI";

 series x=Date y=actual_price / lineattrs=(color=blue) legendlabel="Actual";

series x=Date y=forecast_price / lineattrs=(color=red pattern=dash) legendlabel="Forecast";

title "Nike Price Forecast (Original Scale)";

run;

/* Print the 12-month ahead forecast prices */

proc print data=work.nike_final_forecast noobs;

where Date > '01Jan2024'd;

 var Date actual_price forecast_price lower_price upper_price;

format actual_price forecast_price lower_price upper_price dollar10.2;

title "Nike 12-Month Price Forecast (Actual Dollars)"; run;

/ *Scatter plot of actual prices with a trend line for the forecast */

proc sgplot data=work.nike_final_forecast; where Date >= '01Jan2020'd;

band Upper=upper_price Lower=lower_price x=Date
   / LegendLabel="95% Confidence Limits";

scatter x=Date y=actual_price;

series x=Date y=forecast_price;


run;
```

Happy retirement, Dr. Keating!